MECHENG 306 Design of Sensing and Actuating Systems

Project 2 – Design of Optical Encoders

The Tasks

In a group of 4 you will **design**, **build** and **characterise** two optical encoders:

- Quadrature encoder for the measurement of motor shaft speed and
- Absolute encoder for the measurement of motor shaft position.

You will need to carry out the project using the parts provided over the period of 6 weeks.
You will be assessed by the ***performance*** of the encoders designed/built, the ***code*** enabling the measurements, the ***report*** describing the design and the ***team working*** efforts.

The Challenges

The **Quadrature Encoder** you design/build will measure

(1) the direction of the shaft turning in terms of CW or CCW and
(2) the motor shaft speed in terms of RPM (revolutions per minute).

Notes:

- The maximum shaft speed has been limited to 360 RPM, CW or CCW by the **BaseCodeOne** supplied and the motor power supply setting (6V).
- Only 2 pairs of LED/Phototransistors are allowed to be used for this design.

The **Absolute Encoder** you design/build will measure

(1) the direction of the shaft turning in terms of CW or CCW and
(2) the angular displacement of the shaft turning from any initial position, in terms of degree.

Notes:

- The shaft's angular displacement has been limited to 90 degrees, CW or CCW; and the shaft positioning of a specified angle will repeat 10 times consecutively; each positioning takes 4 sec, which are achieved by the **BaseCodeTwo** provided.
- Only 5 pairs of LED/Phototransistors are allowed to be used for this design.

The Design Kit

A design kit is provided, containing the following components:

Primary

- 1xArduino Uno
- 1xMotor Driver (Pmod HB5)
- 1xDC Motor+Builtin Encoder (IG220019X00015R)
- 16xPhototransistors (940nm, LTR-3208E)
- 10xInfrared LEDs (940nm, TSAL6200)
- 1xSwitching Power Supply (MP-3316)
- 1xDC Power Socket
- 1xBread Board (400 Pin, PSG-BB-400)
- 1xPlastic Box
- 1xTwo-side Tape *(used for trial designs)*
- *1xBlu-Tack (used for trial designs)*
- 1xGlue *(used for the final design)*
- 30xJumper Wires (m to m)30xJumper Wires (f to m)
- 1xUSB Cable

Secondary

- 1xCircular Disk (Acrylic, 10cm in diameter+2mm thickness)

- 1xBase Plate (Acrylic)
- 1xMatrix Plate, Matrix Stand, Base
- 8xSensor Case *(optional to use)*
- 1xMotor Stand
- 4xBolts/Nuts

Notes:

- You are not allowed to use any major parts other than listed in ***Primary***.
- You are allowed to redesign and remake any parts listed in ***Secondary.***
- Other minor parts such as resistors are available in the lab.
- It is not obligatory to use all the parts supplied.
- Data sheets of key parts can be found at Canvas in *Modules/Project 2 Resources/Data Sheets*.

Base Codes

They are available at Canvas in *Modules/Project 2 Resources/Base Codes*.

- **BaseCodeOne**, for the closed-loop control of motor shaft speed, can be used to test and characterise the Quadrature Encoder you will have designed. It will be also used for the assessment.
- **BaseCodeTwo**, for the closed-loop control of motor shaft positioning, can be used to test and characterise the Absolute Encoder you will have designed. It will be also used for the assessment.

Demonstration

When assessed, your Quadrature Encoder will be asked to measure a shaft speed chosen at the time of demonstration, once each for CW and CCW.

- The motor shaft will need to turn for 15 seconds at each commanded RMP, and the shaft speed over the period of 5-10 sec will be assessed. The motor shaft turning this way is achieved by **BaseCodeOne**.
- You will need to display on the monitor (1) CW or CCW, (2) the shaft speeds of both the built-in encoder and your encoder and (3) their difference (i.e., $\Delta\dot{\theta} = \dot{\theta}_{yours} - \dot{\theta}_{builtin}$).
- Your own codes for measurement/display can be added into **BaseCodeOne**.
- The measurement results need to be displayed at the terminal – a sample:

```
RPM from builtin encoder: 99.79
RPM from optical quadrature encoder: 99.37
Error: -0.41
direction read by motor's sensor: CW  ,   direction read by sensor:  CW
```

When assessed, your Absolute Encoder will be asked to measure a shaft position (i.e., angular displacement) chosen at the time of demonstration, once each for CW and CCW.

- Once an angular displacement (e.g., -20 deg; "-" for CCW) is commanded, the motor shaft will be positioned at -20, -40, …, etc at 10 positions consecutively; each positioning takes 4 sec, which is achieved by **BaseCodeTwo** provided.
- You will need to display on the monitor (1) CW or CCW, (2) the positioning angles of both the built-in encoder and your encoder and (3) their difference at each position and the averaged (unsigned) difference over 8 best out of 10 measurements at the end of 10 shaft positioning. The angular difference is defined as $\Delta\theta = \bar{\theta}_{yours} - \bar{\theta}_{builtin}$.
- Your own codes for measurement/display can be added into **BaseCodeTwo**.
- The measurement results need to be displayed at the terminal – a sample:

```
shaft possition from optical absolute sensor from home position: 315.00
shaft displacement from optical absolute sensor: 33.75
Shaft displacement from motor's builtin encoder: 30.00
Error :3.75
```

Please add cw or ccw in your codes!

```
shaft possition from optical absolute sensor from home position: 292.50
shaft displacement from optical absolute sensor: 22.50
Shaft displacement from motor's builtin encoder: 25.26
Error :-2.76

shaft possition from optical absolute sensor from home position: 270.00
shaft displacement from optical absolute sensor: 22.50
Shaft displacement from motor's builtin encoder: 23.68
Error :-1.18
```

Project Report

Suggested table of contents:

Chapter 1 Absolute Encoder Design, by student 1's full names
(sensing circuit design, signal processing, measurement calculation, coding etc)
Chapter 2 Absolute Encoder Performance, by student 2's full names
(disk pattern design and analysis, sensor testing and results, sensor performance, conclusion, suggestions for design improvement)
Chapter 3 Quadrature Encoder Design, by student 3's full name
(sensing circuit design, disk design, signal processing, measurement calculation, coding etc)
Chapter 4 Quadrature Encoder Performance, by student 4's full names
(disk pattern design and analysis, sensor testing and results, sensor performance and analysis, conclusion, suggestions for design improvement)

Notes:

- Each chapter is in no more than TWO pages, single line spacing and 11 size of font of Arial.
- Each chapter is independent, being self-contained of key essential contents and can also refer to other chapters for various purposes.
- Introduction, Conclusion and Appendix are not required and not marked if provided.

Typical issues with technical chapters in the past:

- No schematic diagram of circuitry provided, rather the screenshot of Arduino/breadboard.
- Sensor performance such as accuracy, precision etc characterised only at a single point, rather multiple points evenly distributed within the range.
- No signal output eg waveforms as evidence for signal processing was provided, and discussed.
- Measurement calculations not given in the text rather hidden in codes.
- How did you come up with 5.6M ohm? What about another resistor in series with LED? Which pins of Arduino exactly?
- How did you come up with these values of the two resistors? Are those values same for all 5 circuits?
- No evidence how the performance was characterised, in diagrams? No input-output relation, linearity etc provided.
- Where were the two LED/PT pairs mounted against the tracks, which shows the phase difference?
- The figures are hardly readable.

Assessment Breakdown

(1) Physical Encoders (in group), 55%

- *Quadrature Encoder's performance, 25%*
- *Absolute Encoder's performance, 25%*
- *The codes, 5%*

(2) Report, 40%
- *One technical chapter per student*

(3) Weekly Interview, 5%
- *10 min per group, individually marked, during 2nd weekly lab session*

(4) Peer Evaluation, refer to the template provided at Canvas in *Modules/Project 2 Resources*
- *Adjusting all group marks*

Marking Rubrics for Physical Encoders

(1) Quadrature encoder (25%)

10 points: Both CW and CCW positioning: correct direction and error within ± 2.5 RPM
9 points: Both CW and CCW positioning: wrong direction and error within $\pm$ 2.5 RPM
8 points: Both CW and CCW positioning: correct direction and error within ± 5 RPM
7 points: Both CW and CCW positioning: wrong direction and error within ± 5 RPM
6 points: Both CW and CCW positioning: correct direction and error > ± 5 RPM
5 points: Both CW and CCW positioning: wrong direction and error >± 5 RPM
4 points: Awarded only to those whose demonstration shows evidence of effort, but the sensor fails working properly.
0 point: No show.

(2) Absolute encoder (25%)

10 points: Both CW and CCW positioning: correct direction and error within $\pm$14 degrees
9 points: Both CW and CCW positioning: wrong direction and error within $\pm$ 14 degrees
8 points: Both CW and CCW positioning: correct direction and error within $\pm$ 26 degrees
7 points: Both CW and CCW positioning: wrong direction and error within $\pm$ 26 degrees
6 points: Both CW and CCW positioning: correct direction and error > ± 26 degrees
5 points: Both CW and CCW positioning: wrong direction and error >± 26 degrees
4 points: Awarded only to those whose demonstration shows evidence of effort, but the sensor fails working properly.
0 point: No show.

(3) The code (5%)

(in terms of structure, comment, readability, effectiveness, logic, clarity etc.)
9-10 points (excellent)
7-8 points (good)
5-6 points (satisfactory)
4 base points (awarded to those who have submitted the code)
0 point (no submission)

(4) Individual chapter (30%)

(in terms of technical contents, eg method, math, symbols, description, assumption; visual aid, eg figure, pseudo codes, flowchart, diagram, images; and formatting and language, eg grammar, font, logic, pagination.)
9-10 points (excellent)
7-8 points (good)
5-6 points (satisfactory)
4 base points (awarded to those who have submitted the code)
0 point (no submission)

Demonstration & Submission Dates

- Demonstration in group, 23 Oct, Friday, 1**2:00-2:00** pm, Week 12, Lab 405-559 &405.569
- The Codes (in ino), 25 Oct, Sunday, 10:00 pm, one submission per group
- Report (in PDF), 25 Oct, Sunday, 10:00pm, one submission per group

- Peer evaluation (in PDF), 25 Oct, Sunday, 10:00, one per person

Note:

- The report and codes, one submission per group can be submitted by anyone in the group.